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13. ABSTRACT

Describes a method for evaluation of field heating and cooking equipment operational and functional performance characteristics. Identifies supporting tests, facilities, and equipment required. Provides procedures for adjustment, control accuracy, heat distribution, and efficiency. Not applicable to space heaters, field mess equipment, and test at climatic test sites.

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| 14. KEY WORDS | LINK A | | LINK B | | LINK C | |
|---|---|---|---|---|---|---|
| | ROLE | WT | ROLE | WT | ROLE | WT |
| Field Mess Equipment | | | | | | |
| Field Cooking Equipment | | | | | | |
| Cooking Equipment | | | | | | |
| Food Preparation Equipment | | | | | | |



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AMSTE-RP-702-109
*Test Operations Procedure 10-2-036 1 May 1972
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# FIELD HEATING AND COOKING EQUIPMENT


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## SECTION I
## GENERAL

1. Purpose and Scope. This TOP describes test procedures for evaluating the operational and performance characteristics of field heating and cooking equipment. Equipment covered includes: solid fuel tablets, one and two burner gasoline stoves, five man cooking kit, small detachment field cooking outfit, immersion heater, and liquid fuel fired field range and accessory outfit. Excluded are space heaters and field mess equipment. From the tests listed in Section II, the test director can select those that will satisfy the requirements for the particular test item and the particular test type (i.e., engineering test, initial production test, etc.). This document provides for simulated environmental testing but does not include service testing or environmental testing at climatic test sites.

2. Background. To feed a modern Army and provide needed nutrition is a task requiring high quality food and exacting standards for the equipment used in its preparation. To support an Army at a large installation is task enough, but when the Army moves to the field, the heating and cooking equipment required for food preparation under field conditions must

*This TOP supercedes MTPs 10-2-036 (24 Oct 1969), 10-2-037 (28 Oct 69) and 10-2-040 (19 Dec 67), including all changes.

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be even more carefully designed to ensure efficient operation, ease of cleaning to meet field sanitation conditions and minimum size and weight for a high degree of portability. Safety of operation must also be given more than the usual amount of consideration due to the inherent dangers that exist in small, portable, liquid fuel burning stoves. Field heating and cooking equipment may take the form of a solid chemical tablet for heating water or an individual meal, a one or two burner gasoline stove, a cook kit for five men,a small detachment field cooking outfit or a liquid fuel fired field range and accessory outfit for a platoon that may be employed in multiples thereof to prepare hot food for larger units.

3. Equipment and Facilities. In addition to the equipment and facilities defined in the documents listed in Section II, a standard calorimeter is required to perform the procedures defined by paragraph 7.

## SECTION II
## TEST PROCEDURES

4. Supporting Tests. Common Engineering MTPs/TOPs, Military Standards, the tests defined in Section III, and other published documents to be considered in formulating a test plan are as follows:

| TEST SUBJECT TITLE | PUBLICATION NO. |
|---|---|
| a. Pre-operational Inspection | 10-3-500 |
| (1) Operator Training and Familiarization | 10-2-501 |
| (2) Photographic Coverage | 7-3-519 |
| b. Physical Characteristics | 10-2-500 |
| (1) Plastic Indicator Knobs | MIL-S-10736F Para 4.4.1 |
| (2) Valve and Orifice Hardness | FED-STD-151B Method 243 |
| c. Safety | 10-2-508 |
| d. Performance Tests | |
| (1) Leakage | MIL-C-1588D Para 4.4.1 |
| (a) Field Cooking outfit | Paras 4.4.2; 4.4.4 |
| (b) Single burner stove - tank | |
| - complete stove | |



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| TEST SUBJECT TITLE | PUBLICATION NO. |
|---|---|
| (c) Field range fire unit | MIL-B-40098D |
| -tank assy. | Para 4.4.1 |
| -double tank assy. | Para 4.4.3 |
| -burner unit | Para 4.4.4 |
| -generator | Para 4.4.5 |
| (d) Two burner stove | MIL-S-40608C Para 4.4.2 |
| (2) Ignition and Flame Characteristics | |
| (a) Small detachment cooking outfit | MIL-C-1588D Para 4.4.3 |
| (b) Single burner stove | MIL-S-10736F Para 4.4.5 |
| (c) Field range fire unit | MIL-B-40098D Para 4.4.7 |
| (d) Two burner stove | MIL-S-40608C Para 4.4.3 |
| (3) Controls and Adjustments (Refer to para 5) | |
| (4) Heat Distribution | |
| (a) Stoves and griddles (Refer to para 6) | |
| (b) Stove/Range Ovens | MIL-B-12570C Para 4.4.5 |
| (5) Efficiency (Refer to para 7) | |
| (6) Solid Fuel Packet | MIL-F-10805C Amend. -1 Paras 4.3.2; 4.4 |
| (7) Immersion Heaters | MIL-H-1597B Para 4.4 |
| e. Environmental Testing | |
| (1) Altitude | MIL-STD-810B Method 500 |
| (2) Temperature | Method 501 |
| (3) Sunshine | 4-2-826 |
| (4) Rain | 2-2-815 |
| (5) Humidity | 4-2-820 |
| (6) Fungus | 4-2-818 |
| (7) Salt Fog | MIL-STD-810B Method 509 |
| (8) Dust Test | Method 510 |
| (9) Vibration | 4-2-804 |
| (10) Rough Handling | 4-2-602 |



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| | TEST SUBJECT TITLE | PUBLICATION NO. |
|---|---|---|
| f. | Surface Transportability (General Supplies and Equipment) | 10-2-503 |
| g. | Human Factors Evaluation | 10-2-505 |
| h. | Reliability<br>Confidence Intervals and Sample Size | AMCP 702-3<br>3-1-002 |
| i. | Endurance Testing | 10-2-502 |
| j. | Maintenance Evaluation | 10-2-507 |
| k. | Value Analysis | USAMC SUPPL 1 to AR 11-26 |

## SECTION III
## SUPPLEMENTARY INSTRUCTIONS

5. Controls and Adjustments.

a. Objective. To determine the accuracy of operation of test item controls and adjustments.

b. Method. The test item is placed in an area free of air currents and external conditions that would affect the validity of this test. The fuel supply and metering valve controls (individual or combined) are turned on and the test item ignited. The controls are then adjusted for maximum output and the height of the flame and rate of fuel flow are determined. The controls are then adjusted for three-quarters, one-half, and one-quarter maximum output, and minimum output. Pertinent measurements at each setting are recorded.

c. Data Required.

(1) Nomenclature and type of test item.

(2) Flame heights and rate of fuel flow determined at maximum, three-quarters, one-half and one-quarter of maximum, and minimum outputs indicated by control settings.

d. Analytical Plan. The flame heights and rates of fuel flow are tabulated by control settings and the tabulations analyzed to determine control setting accuracy. The results of the analysis are compared to the requirements of the MN to determine conformance to specifications.



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6. Heat Distribution.

a. Objective. To determine the heat distribution of the test item.

b. Method. (Applicable to stoves and field ranges.) The test item is prepared for testing by placing the specified type and size utensil for frying over the heat source. The utensil is instrumented by five remote reading temperature devices, one placed near each corner and one in the center. The test item is ignited and adjusted for a normal operating temperature. When the temperature is stablized, the five thermocouple indications are recorded simultaneously by location. The procedure is repeated for various indicated temperatures as required.

c. Required Data.

(1) Nomenclature and type of test item.

(2) Description of utensil used.

(3) Sketch of thermocouple locations.

(4) Tabulation of simultaneous readings by location, on utensil for each heat setting of burner unit.

d. Analytical Plan. An analysis is made of the difference in temperature between each point. The temperatures of the hottest and coldest of the five points are determined. This difference is compared with the requirements of the MN to determine conformance to specifications.

7. Efficiency.

a. Objective. To determine the operating efficiency of the test item.

b. Method. The test item is instrumented with an accurate fuel input measuring device and a calorimeter to record heat output. The burner is ignited and adjusted to stablized operation for maximum heat output. This condition is maintained for a period of one hour, during which time interval the quantity of fuel consumed and the heat output in BTU are accurately determined. This procedure is repeated for one-half maximum and minimum heat output settings. During the above procedures, the temperature of the flue gas is determined and a sample obtained for analysis.



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c. Required Data.

(1) Nomenclature and type of test item.

(2) Type of fuel and published heating value in B.T.U.

(3) Fuel consumed during each test period (to the nearest tenth of a gallon).

(4) Heat output measured during each test period.

(5) Ambient temperature recorded for each test.

(6) Flue gas temperature and analysis during each test period.

d. Analytical Plan. The measured heat output during each test period is divided by the heating value of the fuel consumed and the result multiplied by 100 to obtain the efficiency of operation. The efficiencies of each test are averaged to obtain the average overall operating efficiency of the test item which, together with the $CO_2$ content of the flue gas is compared with the requirements of the MN to determine conformance to specifications.

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# APPENDIX
# REFERENCES

1. AR 70-38, "Research, Development, Test, and Evaluation of Materiel for Extreme Climatic Conditions."

2. USAMC Supplement 1 to AR 11-26, "Value Engineering."

3. AMCP 702-3, "Quality Assurance - Reliability Handbook."

4. Federal Test Method Standard 151B, "Metals, Test Methods", including notice 1.

5. MIL-STDF810B, "Environmental Test Methods", including notices 1 thru 4.

6. MIL-C-1588D, "Cooking Outfit, Field, Small Detachment."

7. MIL-H-1597B, "Heater, Immersion, Liquid Fuel Fired, for Cans, Corrigated."

8. MIL-S-10736F, "Stove, Gasoline Burner, M1950, With or Without Case", including amendments 1 and 2.

9. MIL-F-10805C, "Fuel, Compressed, Trioxane, Ration Heating", including amendment 1.

10. MIL-B-12570C, "Bakery Oven, Trailer Mounted, M-533", including amendment 1.

11. MIL-B-0040098D, "Burner Unit, Gasoline, Field Range Outfit, M2."

12. MIL-S-40608C, "Stove, Gasoline, 2-Burner."



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